

Evolving past instabilities on the thermally pulsing-(super)asymptotic giant branch

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ABSTRACT

We address the challenge of running thermally pulsing-(super)asymptotic giant branch [TP-(S)AGB] models, with a 1D hydrostatic stellar evolution code, without suffering instabilities that terminate the evolution. We investigate two instabilities that usually occur during the luminosity peak following a thermal pulse: the hydrogen recombination instability and the Fe-peak instability. Both instabilities occur when the stellar mass is significantly reduced ($M \lesssim M_i/2$) at the end of the TP-(S)AGB in our models with initial mass $M_i \gtrsim 2 M_\odot$. The hydrogen recombination instability occurs due to the difficulty of modelling a thermally and dynamically unstable envelope in a 1D hydrostatic code, and is prevented by damping the energy released by hydrogen recombination in the outer envelope. The Fe-peak instability occurs when the radiation pressure drops at the base of the convective envelope and is prevented by boosting the convective energy transport in this region. We provide custom routines to prevent these instabilities in the stellar evolution code MESA. The impact of these routines on the stellar structure is minimized so as to not affect the efficiency of third dredge-up, hot-bottom burning, or the wind mass-loss rate. We find only a modest reduction in third dredge-up efficiency at small envelope masses ($M_{\text{env}} \lesssim 1.0 M_\odot$). Consequently, our $M_i = 5 M_\odot$ star, with hot-bottom burning, becomes a carbon star for the last ~ 10 per cent of its thermally pulsing lifetime. The largest stellar radii are reached during the final thermal pulses, which may have important consequences for binary–star interactions.

Key words: stars: AGB and post-AGB – carbon – evolution.

1 INTRODUCTION

Stars massive enough to ignite helium ($M \gtrsim 0.8 M_\odot$), but not massive enough to reach the conditions for carbon burning ($M \lesssim 7 M_\odot$), reach the asymptotic giant branch (AGB) as they near the ends of their lives (Iben & Renzini 1983; Busso, Gallino & Wasserburg 1999; Herwig 2005; Karakas & Lattanzio 2014). Similarly, stars that are massive enough to ignite carbon but not further burning stages ($7 \lesssim M/M_\odot \lesssim 10$) reach the superasymptotic giant branch (SAGB) where they form oxygen–neon cores (Garcia-Berro & Iben 1994; Siess 2006, 2007, 2010; Doherty et al. 2010, 2014, 2015; Farmer, Fields & Timmes 2015). During the early-AGB, the helium-burning shell moves outwards in mass until the helium-rich intershell region is reduced to $\lesssim 0.001\text{--}0.1 M_\odot$. The star then reaches the thermally pulsing-(S)AGB [TP-(S)AGB].

During a thermal pulse, the helium-burning shell ignites in a thermonuclear runaway known as the He-shell flash. A temporary convective region forms in the intershell, known as the pulse-driven convection zone (PDCZ), which mixes carbon, and some oxygen, from the bottom of the intershell throughout the region (Boothroyd & Sackmann 1988; Herwig 2000; Lugaro et al. 2003). The energy produced by the He-shell flash then causes a rapid expansion of the stellar envelope that temporarily extinguishes the hydrogen-burning shell. The convective envelope extends further towards the core and,

if the flash is strong enough, it reaches down into the intershell region in a third dredge-up (TDU) event. The TDU mixes CO-enhanced material into the envelope (Iben & Renzini 1981, 1983). The material has a higher abundance of carbon than oxygen and thus the ratio of carbon to oxygen atoms, C/O, increases. If C/O exceeds unity, the star transitions to a carbon star (Groenewegen, van den Hoek & de Jong 1995; Wallerstein & Knapp 1998; Marigo et al. 2020; Unnikrishnan et al. 2023). Once the energy produced by the He-shell flash is radiated away, the envelope contracts again and the hydrogen-burning shell re-ignites. The intershell mass is increased by the hydrogen-burning shell until conditions are reached once again for the He-shell flash. This process of alternative hydrogen- and helium-shell burning results in periodic thermal pulses that only end once the envelope is lost due to the strong stellar wind (Bedijn 1988; Vassiliadis & Wood 1993; Bloeker 1995; Höfner & Freytag 2020). The star then evolves along the post-AGB before cooling as a CO or ONe white dwarf (WD).

TP-(S)AGB stars are important contributors to galactic chemical evolution. First, their surfaces can be enriched in carbon and oxygen by TDU events. Secondly, hot-bottom burning (HBB) in stars with $M \gtrsim 4 M_\odot$ converts carbon into nitrogen in the envelope (Scalo, Despain & Ulrich 1975; Lattanzio 1992) and results in nitrogen-rich stars (Wood, Bessell & Fox 1983; McSaveney et al. 2007; Pols et al. 2012; Fernández-Trincado et al. 2020). Finally, TP-AGB stars are thought to be the main contributors to the production of s-process elements (Travaglio et al. 2001; Straniero, Gallino & Cristallo 2006;

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Kobayashi, Karakas & Lugaro 2020; Prantzos et al. 2020). s-process elements are produced via the formation of a ^{13}C pocket in the intershell region during TDU (Cameron 1957; Smith & Lambert 1986; Busso et al. 2001; Lugaro et al. 2003; Goriely & Siess 2004). Once produced, s-process elements are mixed throughout the intershell region by the PDCZ and then mixed into the stellar envelope during the next TDU event.

Despite their importance, much of the physics of AGB stars is not well constrained. This is due to the difficulty in both producing accurate theoretical models and obtaining observations of AGB and post-AGB stars. Two of the main uncertainties in AGB evolution are convection and mass-loss (Höfner 2016). The future of AGB modelling is pointed towards combining the results of 3D hydrodynamical models and 1D hydrostatic models (Herwig 2008). For example, results from 2D and 3D hydrodynamical models are beginning to be used to calibrate the mixing length theory (MLT) parameters in 1D models (e.g. Freytag, Höfner & Liljegren 2018; Sonoi et al. 2019; Higl, Müller & Weiss 2021; Goldberg, Jiang & Bildsten 2022). In the absence of such calibrations from multidimensional hydrodynamical models, observations can be used instead to validate the results of 1D models (e.g. Pastorelli et al. 2019, 2020; Marigo 2022). Similarly, population synthesis codes that implement the results of 1D models can be used for comparison to observed populations, including binaries (e.g. Izzard & Tout 2004; Izzard et al. 2004). Thus, to constrain AGB physics we need to be able to reliably run grids of 1D stellar evolution models.

This work addresses the issue of instabilities and convergence issues on the TP-AGB that prevent the production of 1D models that evolve all the way to the end of the TP-AGB and make the running of large grids of models time-consuming. In particular, we focus on producing complete AGB models using the stellar evolution code MESA; however, many of the problems and methods discussed are applicable to other full stellar evolution codes. The onset of instabilities is investigated in detail and methods of prevention of such instabilities provided. The produced methods are general such that they are applicable to all masses and do not depend on fixed physical assumptions such as convective overshooting or wind mass-loss. This work focuses on solar metallicity ($Z = 0.014$) models; however, the methods should easily extend to other metallicities.

In Appendix A, we cover controls specific to MESA required to prevent convergence problems during the He-shell flash. In Section 2, we highlight some important modelling assumptions. We then investigate two instabilities that occur in our TP-(S)AGB models, the hydrogen recombination instability (HRI) (Section 3), and the Fe-peak instability (FeI) (Section 4). Finally, in Section 5 we discuss the extent to which instabilities plague other stellar evolution codes and the important consequences of being able to model the final thermal pulses that occur at low envelope mass.

2 STELLAR MODELLING

For all computation, we use version 23.05.1 of the 1D stellar evolution code MESA (Paxton et al. 2011, 2013, 2015, 2018, 2019; Jermyn et al. 2022). Full MESA controls, including `inlists` and `run_star_extras` code, are available on the MESA marketplace.¹ We use the models produced in Rees, Izzard & Karakas (2024, hereafter Paper I) at the onset of the first thermal pulse as a starting point for TP-(S)AGB computations in this work. We also make the same choices for stellar physics as Paper I. Note that an older version

of MESA (15140) was used in Paper I; however, for this work we use an updated version that utilizes `auto_diff` functionality (Jermyn et al. 2022). We do not find that the update in version makes a difference to the TP-(S)AGB evolution but it does allow for easier implementation of custom routines. To ensure that TDU events are resolved, we use the resolution routine developed in Paper I as well as the method to prevent runaway TDU in massive and super-AGB models that experience hot TDU. Appendix B contains some further discussion around resolving the TDU at reduced envelope masses. While the treatment of convection and stellar winds is discussed in detail in Paper I, in this work we discuss in greater detail the choice of mixing length parameter, a required parameter in the MLT of convection (Böhm-Vitense 1958; Henyey, Vardya & Bodenheimer 1965; Joyce & Tayar 2023).

2.1 Mixing length parameter

The mixing length parameter, α_{mlt} , is often constrained in stellar evolution models using the position of the Sun in the Hertzsprung-Russell (HR) diagram. However, there is no reason to believe that the same value should be correct for different stellar masses or different evolutionary stages. Indeed, the mixing length parameter during the AGB is particularly badly constrained. 3D simulations have been used to constrain the mixing length parameter in low-mass stars up the giant branch (Sonoi et al. 2019; Spada, Demarque & Kupka 2021). However, such models do not extend to the AGB due to additional complications relating to the increased size of convective cells (Freytag et al. 2018). In the absence of such constraints, we use $\alpha_{\text{mlt}} = 1.931$ (Cinquegrana & Joyce 2022). However, there is some evidence for higher values of the mixing length parameter during the AGB phase (Porter & Woodward 2000; McSaveney et al. 2007). In addition, the mixing length parameter is thought to reach values as high as $\alpha_{\text{mlt}} \approx 3\text{--}4$ in simulations of red supergiants (Goldberg et al. 2022).

To test the impact of the mixing length parameter, we compute the TP-AGB phase of an $M_i = 3 M_\odot$ star with $\alpha_{\text{mlt}} = 1.931$ and compare to a model sequence with an increased value of $\alpha_{\text{mlt}} = 2.5$ (Fig. 1). We also tested an increased mixing length parameter just in the PDCZ, but we found this to have no impact on the energy released by He-shell flashes or consequent evolution. This is unsurprising because in the deep stellar interior convection is very efficient and the temperature gradient is nearly adiabatic (Woodward, Herwig & Lin 2015). Thus, the mixing length parameter should not have an impact in the PDCZ. As previously found, an increased mixing length parameter in the convective envelope leads to deeper TDU (Herwig 2008; Herwig et al. 2012). The increased mixing length parameter also compresses the envelope, thus decreasing the stellar radius. Because our mass-loss prescription is dependent on the stellar radius, this causes there to be many more thermal pulses before the model star enters the superwind phase. The increased dredge-up efficiency and longer thermally pulsing lifetime both increase enrichment in nuclear processed material and thus strongly impact the stellar yields of intermediate-mass stars (Karakas & Lattanzio 2007; Cristallo et al. 2009, 2011, 2015; Karakas 2010; Karakas & Lugaro 2016; Pignatari et al. 2016; Ritter et al. 2018).

In SAGB models, the situation differs. Increasing α_{mlt} increases the temperature at the base of the convective envelope and thus increases the luminosity produced by HBB (Doherty et al. 2014). Because these stars are always in the superwind phase of the Vassiliadis & Wood (1993) wind mass-loss, this leads to higher mass-loss rates and fewer thermal pulses. Doherty et al. (2014) also found no increase in the efficiency of TDU with larger values of α_{mlt} .

¹https://cococubed.com/mesa_market/inlists.html

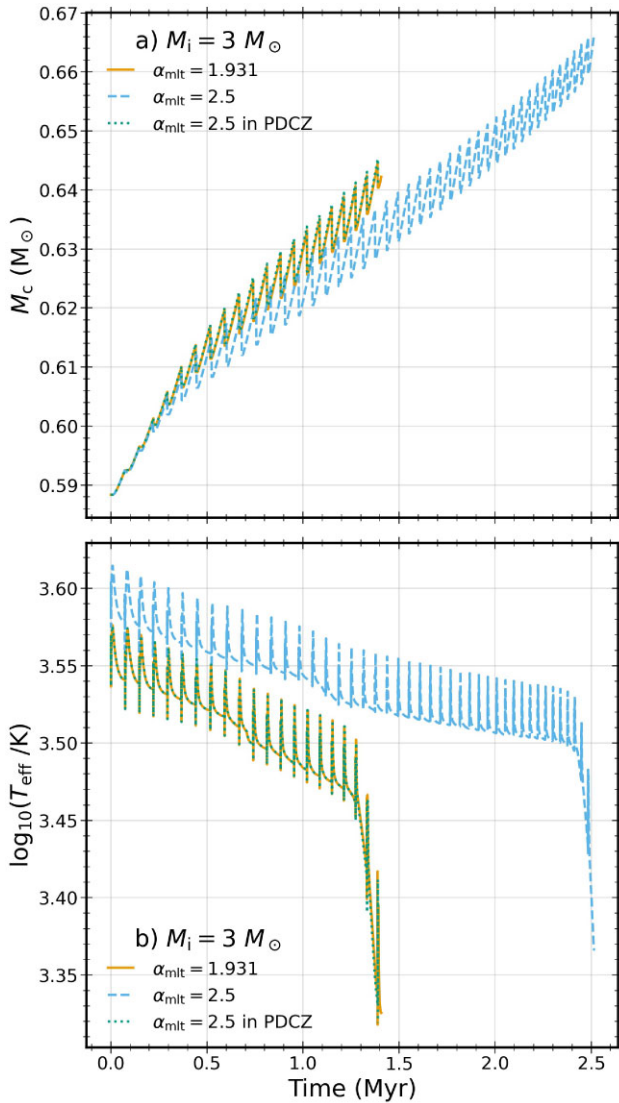


Figure 1. Thermal pulses for a $M_i = 3 M_\odot$ model star at solar metallicity when the mixing length parameter, α_{mlt} , is chosen to be 1.931 (solid line) versus 2.5 (dashed line). We also include a model sequence in which α_{mlt} is increased to 2.5 only in the PDCZ (dotted line); however, this follows to the same evolution as the sequence with $\alpha_{\text{mlt}} = 1.931$ everywhere. (a) The core mass evolution reveals more efficient TDU for the larger mixing length parameter. (b) The effective temperature is higher for the more compressed star with the larger mixing length parameter.

3 HYDROGEN RECOMBINATION INSTABILITY

TP-(S)AGB stars have cool, expanded envelopes ($T_{\text{eff}} \lesssim 5000$ K), which are susceptible to exponentially growing envelope pulsations driven by the ionization and recombination of hydrogen with a supporting effect from helium (Wood 1974; Tuchman, Sack & Barkat 1978, 1979; Clayton 2018; Trabucchi et al. 2019; Gautschi 2023). As the star evolves along the TP-(S)AGB and the envelope mass decreases, the period and growth rate of these pulsations both increase (Trabucchi et al. 2019). In a hydrostatic code, such pulsations should be smoothed through, however the envelope is still unstable. Wagenhuber & Weiss (1994, hereafter WW94) investigated an instability leading to a runaway expansion during the luminosity peak of a thermal pulse. They found the instability is driven by the recombination of hydrogen within a ‘thermally and dynamically

unstable’ envelope and thus named it the HRI. The energy released by recombination of hydrogen leads to further expansion and cooling, which in turn leads to further recombination causing a runaway. Gautschi (2023) discuss that in the envelopes of very luminous AGB stars, the thermal and dynamical time-scales become comparable and propose that the HRI is a result of envelope pulsations in thermal strange modes.

There are similarities between the HRI and the contribution of hydrogen recombination to the envelope ejection of a common envelope event (Ivanova 2018). However, the existence of the HRI in 1D hydrostatic models is not evidence for the large scale expansion and ejection of the envelope of TP-(S)AGB stars. In more precise calculations, such as 3D hydrodynamic models, envelope pulsations dominate, which are known to elevate material sufficiently for it to be lost in a stellar wind (Bedijn 1988; Höfner & Olofsson 2018; Höfner & Freytag 2020; Freytag & Höfner 2023). Pulsations can also lead to dynamical mass ejections (Tuchman et al. 1978; Clayton 2018). Thus, modelling the envelope pulsations is essential to constrain wind mass-loss. 1D hydrostatic models however are still very useful for modelling the stellar interior where pulsations do not dominate.

3.1 Onset of the HRI

Fig. 2(a) shows the occurrence of the HRI terminating the evolution of a model with $M_i = 3 M_\odot$. The rapid expansion is accompanied by a rapid decrease in effective temperature. However, WW94 found that if the spatial resolution is insufficient, the HRI can manifest in either a decreasing or increasing or an alternating divergence of the surface properties. Similarly, if the temporal resolution is low, the HRI may be smoothed over and can be avoided for a number of thermal pulses.

The HRI terminates our (S)AGB models with $M_i \gtrsim 2 M_\odot$. The remaining envelope mass and T_{eff} at the onset of the HRI for each model is detailed in Table 1. The HRI occurs when the envelope mass is significantly reduced ($M \lesssim M_i/2$) and the envelope has expanded and cooled in response to mass-loss. Our low-mass models ($M_i \lesssim 4.5 M_\odot$) are also significantly enhanced in carbon ($\text{C/O} > 1$). This results in a higher opacity and further expansion and cooling of the envelope. Our results differ from those of WW94 in the amount of mass-loss required before the HRI occurs. They found the HRI to occur after only $\sim 0.1 M_\odot$ of mass was lost in models with $2 \leq M_i / M_\odot \leq 4$. However, the effective temperatures at the onset of the HRI were in the range $T_{\text{eff}} \sim 2000\text{--}2200$ K, which is somewhat lower than the range found in our models, $T_{\text{eff}} \sim 2200\text{--}2500$ K, for the same mass range. This difference is likely due to various assumptions in physics that resulted in their model envelopes being cooler, even at higher envelope mass, and thus more susceptible to the HRI.

Since WW94 there has not been much mention of the HRI, although similar behaviour is found in the models of Bertolami (2016). WW94 postulate that if a stellar evolution model is accurate enough, the HRI is inevitable. This is due to the inability of a hydrostatic model to properly account for the energy released by hydrogen recombination that, in a hydrodynamic model, drives envelope pulsations. Due to the sensitivity of the HRI to the time-steps taken, the HRI may be smoothed over by crude time resolution. In addition, different assumptions of physics can result in higher effective temperatures and hence less unstable envelopes. Combined with a crude enough time resolution, the HRI could be avoided in such models. Many researchers may have also encountered the HRI but not recognized it as such. The impact of the HRI on the stellar surface properties can be similar to the FeI (Section 4); however, it is

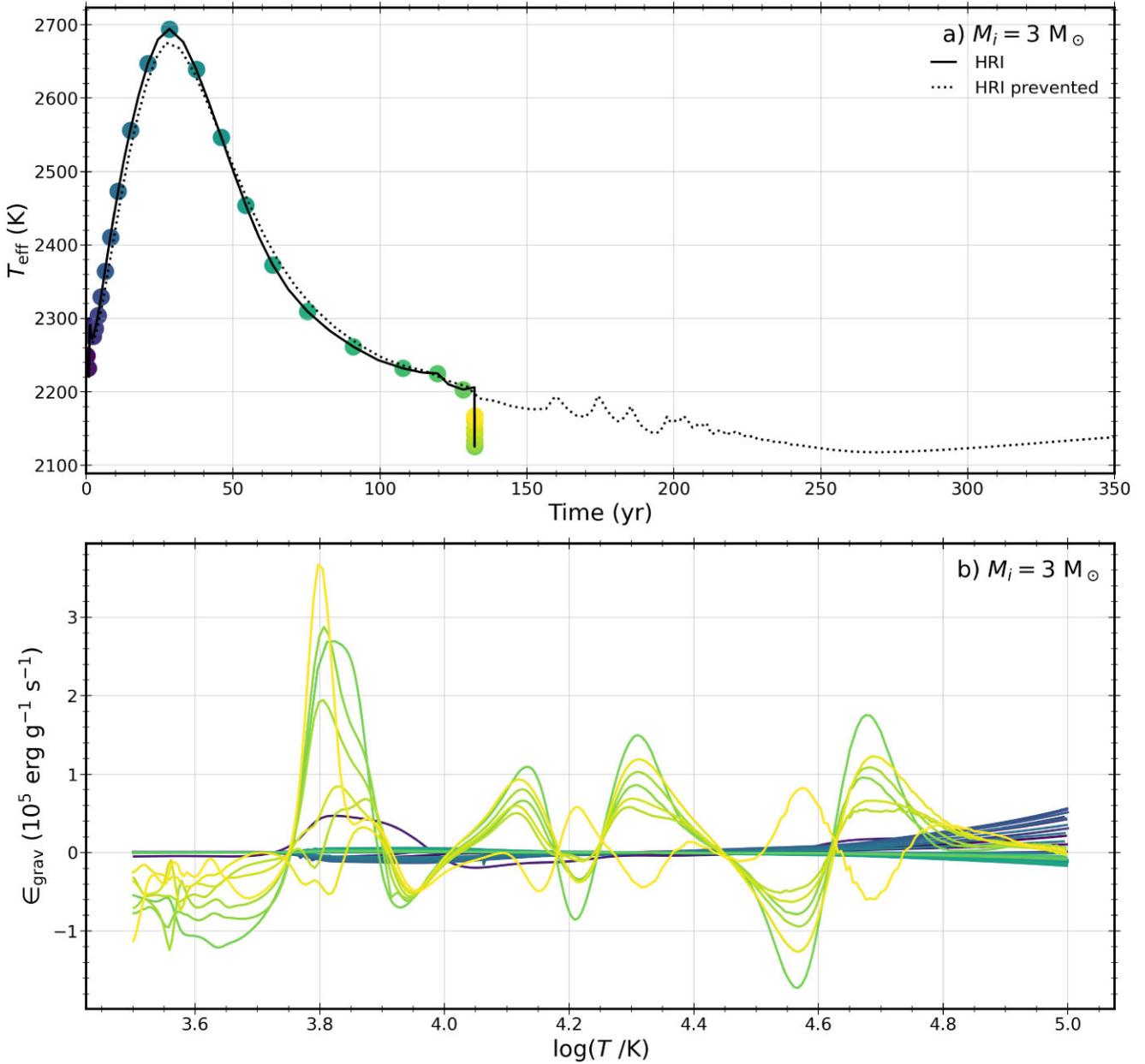


Figure 2. The onset of the HRI terminating the evolution of an $M_i = 3 M_\odot$, solar metallicity TP-AGB model star during the 22nd thermal pulse. (a) The HRI manifests as a sudden change in surface properties due to the rapid recombination of hydrogen (solid line). The HRI can be prevented as described in Section 3.2 (dotted line). (b) The energy produced/absorbed by recombination/ionization during the HRI builds up to extreme spatial oscillations in the outer envelope. The colours of the scatter points in panel a are the same as the corresponding energy profiles in panel (b).

not accompanied by the decrease in gas to total pressure ratio which identifies the FeI. For example, Karakas & Lattanzio (2007) detail a convergence problem, associated with a rapid increase in the stellar radius, that occurs after the He-shell flash but is not associated with small values of the gas pressure to total pressure. We postulate that this was likely to be the HRI.

3.2 Preventing the HRI

The HRI occurs when the star has lost a significant amount of mass and is near the end of its nuclear-burning lifetime. At this point there are large and frequent mass ejections triggered by pulsations

(Clayton 2018), which is mimicked by the superwind phase of the Vassiliadis & Wood (1993) mass-loss rate. However, we do wish to finish the evolution along the TP-AGB to have complete model sequences and a starting point for the post-AGB phase. Thus, we wish to damp out energy injections from hydrogen recombination to prevent the HRI from transpiring. In this way we can continue to compute a hydrostatic equilibrium model and rely on our imposed mass-loss prescription to evolve on to the post-AGB. In MESA, energy contributions from ionization and recombination are included in the ϵ_{grav} term of the energy equation. ϵ_{grav} is calculated from the rate of change of the internal energy and density. Further details of this quantity and the energy equation are discussed in Appendix C. In

Table 1. The core mass, M_c , envelope mass, $M_{\text{env,HRI}}$, thermal pulse number, $N_{\text{TP,HRI}}$, and effective temperature, $T_{\text{eff,HRI}}$ at the onset of the HRI in models with initial masses $2.0 \leq M_i / M_\odot \leq 8$.

$M_i (M_\odot)$	$M_c (M_\odot)$	$M_{\text{env,HRI}} (M_\odot)$	$N_{\text{TP,HRI}}$	$T_{\text{eff,HRI}} (K)$
2.00	0.61	0.10	19	2600
2.25	0.61	0.11	24	2400
2.50	0.61	0.71	23	2500
2.75	0.63	0.62	22	2200
3.00	0.64	0.61	22	2200
3.25	0.67	0.59	20	2200
3.50	0.71	1.29	18	2500
3.75	0.76	0.69	16	2300
4.00	0.82	1.01	18	2500
4.50	0.85	0.88	22	2600
5.00	0.88	0.82	28	2700
5.50	0.92	0.71	31	2700
6.00	0.96	0.82	34	2700
6.50	1.00	1.54	33	2700
7.00	1.05	1.48	45	2800
7.50	1.12	1.52	75	2900
8.00	1.19	2.62	101	2900

panel b of Fig. 2, we plot ϵ_{grav} in the outer envelope of a $M_i = 3 M_\odot$ model as it changes over a thermal pulse cycle. The colour of profiles in panel b match their corresponding point on the time evolution in panel a. When the HRI occurs, we see large spatial oscillations in energy release, of the order of $10^5 \text{ erg g}^{-1} \text{ s}^{-1}$, due to the unstable recombination and ionization of hydrogen. These cause numerical problems for the solver that eventually grinds to halt. To prevent the HRI, we can force the contribution to the energy equation from ionization and recombination to be zero in the outer envelope where $\log(T / K) < 4.2$. Details for the implementation of this method in MESA are given in Appendix D. For the thermal pulse cycle shown in Fig. 2, this region encompasses the outer $\approx 0.15 M_\odot$ of mass at the point of minimum radius and the outer $\approx 0.4 M_\odot$ at the point of maximum radius. This is between ~ 20 and 50 per cent of the envelope mass, which has been reduced to $\approx 0.73 M_\odot$ by the mass-loss prescription. The dotted line in Fig. 2 (panel a) shows the evolution of effective temperature when the HRI is inhibited using this method. We find that discarding this energy source reduces the peak in T_{eff} by $\approx 20 \text{ K}$ but otherwise has very little impact. This shows that prior to the instability, the recombination/ionization energy has only a small contribution to the total energy production which is dominated by shell burning sources. Thus, discarding this energy source does not affect the stellar evolution, other than by preventing the HRI. However, it is possible that at further reduced envelope masses, there is an increased contribution of the recombination/ionization energy to the expansion and contraction of the envelope due to a greater fraction of the envelope mass being included in the low temperature region. Thus, our approximation may begin to impact the surface properties. However at this point, envelope pulsations will be strongly driven and hydrostatic computations of the surface properties should be taken with a pinch of salt anyway.

3.3 The HRI during the He-shell flash

We find that the HRI not only occurs during the luminosity peak but also during the He-shell flash in models with $M_i \gtrsim 3 M_\odot$. During the He-shell flash, the time-step shortens to resolve the thermonuclear runaway. Such small time-steps facilitate the HRI at higher effective temperatures than are reached during the following luminosity peak.

This can be seen in Fig. 2(b) where there is a peak in ϵ_{grav} around $\log(T / K) \approx 3.9$ during the He-shell flash at the start of the thermal pulse cycle (dark purple line). For this particular thermal pulse, MESA converges without a runaway instability; however, this growth in ϵ_{grav} does lead to a full instability at lower effective temperatures. The HRI during the He-shell flash can be prevented with the same method we use to prevent the HRI during the luminosity peak.

4 FE-PEAK INSTABILITY

The second instability we encounter is the FeI. Lau et al. (2012, hereafter L12) found the FeI to be caused by the local opacity maximum at the bottom of the convective envelope, produced by M-shell electrons of Fe-group elements. The iron opacity peak leads to the accumulation of an energy excess in this region, instigating a situation where the radiation pressure dominates over the gas pressure, and locally the luminosity exceeds the Eddington luminosity. L12 find that the ratio of the gas to total pressure drops and tends to zero and thus the model fails to converge. The FeI has been identified in many different TP-(S)AGB models produced with a range of stellar evolution codes (e.g. Wood & Faulkner 1986; Herwig 2005; Bertolami & Althaus 2006; Doherty et al. 2014; Karakas 2014) as well as post-AGB models (Bertolami 2016; Oomen 2020). Due to the iron opacity peak being the root cause of the instability, it does not occur when older opacity tables, without said peak, are used.

As the instability develops, the model envelope expands and the density at the bottom of the convective envelope drops. Wood & Faulkner (1986) suggested that in a real star, a hydrodynamic expansion of the envelope occurs and thus the convergence problems that develop in stellar models are a consequence of the lack of existence of a hydrostatic solution. It is unclear whether this instability leads to the total ejection of the envelope, however it is likely to lead to significant mass-loss. We cannot, however, use 1D hydrostatic models to constrain the impact of the iron opacity peak on mass-loss as 3D hydrodynamical models are required. Such simulations must couple the behaviour of pulsations and shocks with radiative physics to model the complex processes occurring in the envelope (e.g. Freytag et al. 2018; Freytag & Höfner 2023). Instead, we ‘trust’ our wind mass-loss routines and continue to evolve our first approximation, 1D hydrostatic model past instabilities.

4.1 Onset of the Fe-peak instability in TP-AGB models

As AGB stars evolve, they lose a significant amount of their mass due to strong stellar winds (and potentially binary interactions), causing the density in the envelope to decrease. Consequently the ratio of the gas pressure to the total pressure at the bottom of the convective envelope decreases as the model evolves. Eventually such conditions lead to the onset of the FeI and a failure in convergence. Fig. 3(c) shows $\beta = P_{\text{gas}} / P$ throughout the envelope during the FeI in an $M_i = 4 M_\odot$ model, showing the sharp drop in β at the base of the convective envelope.

L12 found the remaining envelope mass at the onset of the instability is around $1\text{--}2 M_\odot$ in model sequences with initial masses in the range $7\text{--}10 M_\odot$. They also predicted that the instability could occur at initial masses as low as $2.5 M_\odot$ at solar metallicity. We have investigated the occurrence of the FeI in solar metallicity AGB models as shown in Table 2. These models were ran using envelope damping to avoid the HRI, as described in Section 3. Note that this damping stops the FeI developing to the point of convergence failure but instead the time-step is prohibitively small as $\beta \rightarrow 0$. We

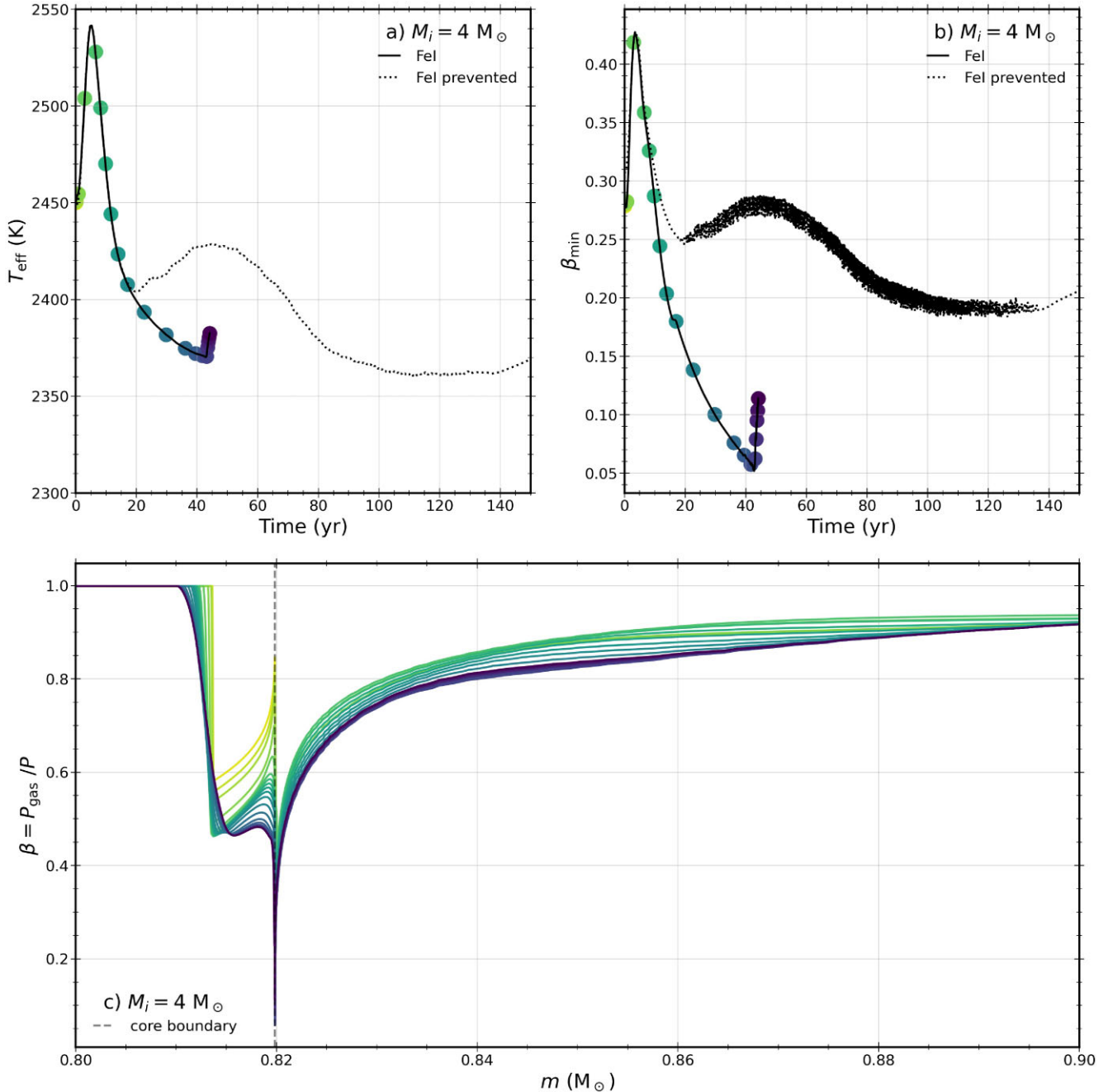


Figure 3. The onset of the FeI terminating the evolution of a $M_i = 4 M_{\odot}$, solar metallicity TP-AGB model star during the 18th thermal pulse. (a) The evolution of the effective temperature up to the occurrence of the FeI (solid line) compared to the evolution when the FeI is prevented with the routine described in Section 4.2 (dotted line). (b) the profile of $\beta = P_{\text{gas}}/P$ throughout the envelope showing the sharp drop in β at the bottom of the convective envelope at around $m \approx 0.82 M_{\odot}$. The colours of the scatter points in panel (a) are the same as the corresponding beta profiles in panel (b). The core boundary is defined to be the outermost location where the ^1H mass fraction is less than 0.01 and the ^4He mass fraction is greater than 0.1.

find that the instability occurs in stars with core masses as low as $0.62 M_{\odot}$. For the 2 and $3 M_{\odot}$ initial masses, the instability occurs as the star evolves off the AGB. In post-AGB models, the FeI does not lead to a total failure of convergence however it does reduce the time-step and thus result in very slow computation of the post-AGB evolution (Bertolami 2016). In all models with $M_i > 3 M_{\odot}$, the instability occurs during the luminosity peak of a thermal pulse, when the envelope expands in response to a He-shell flash. This causes the density at the base of the convective envelope to decrease and thus trigger the instability.

4.2 Preventing the Fe-peak instability

Solutions for preventing the FeI have been discussed in several works (e.g. Lau et al. 2012; Bertolami 2016). In a region where convection is efficient, the local luminosity can exceed the Eddington luminosity without a drop in β . Thus, the FeI can be avoided by increasing the mixing length parameter to increase the efficiency of convection and transport energy out of the problem region. However, increasing the mixing length parameter in the entire envelope affects the envelope structure and thus the subsequent evolution (Section 2.1). We find

Table 2. The thermal pulse number, N_{TP} , core mass, M_c , and envelope mass, M_{env} , at the onset of FeI for solar metallicity models with initial masses in the range $2 \leq M/M_\odot \leq 7$. For the $M_i = 2 M_\odot$ model, the FeI occurs during the interpulse period as the model begins to transition on to the post-AGB. For all higher mass models, the FeI occurs during the luminosity peak of a thermal pulse.

$M_i (M_\odot)$	N_{TP}	$M_c (M_\odot)$	$M_{\text{env,FeI}} (M_\odot)$	Instability
1	7	0.57	—	No instability
2	19	0.62	0.02	End TP-AGB
3	21	0.64	0.32	End TP-AGB
4	18	0.82	0.95	L-peak
5	27	0.88	1.04	L-peak
6	32	0.96	1.20	L-peak
7	47	1.05	1.34	L-peak

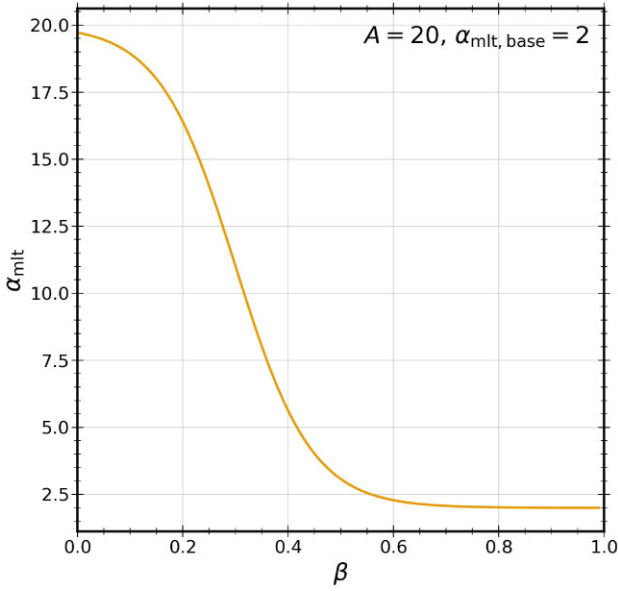


Figure 4. α_{mlt} as a function of β calculated using equation (1) for $A = 20$ and $\alpha_{\text{mlt,base}} = 2.0$. α_{mlt} is increased as β decreases at the base of the convective envelope to boost the efficiency of energy transport and prevent the FeI in TP-(S)AGB stars with reduced envelope mass.

an effective solution to avoid the FeI without changing the envelope structure is to increase the mixing length parameter in only the region at the base of the convective envelope where β is small.

To target the specific region, we increase the mixing length parameter as a function of $\beta = P_{\text{gas}}/P$, such that

$$\alpha_{\text{mlt}} = A - \frac{A - \alpha_{\text{mlt,base}}}{1 + 10^{-6(\beta-0.3)}}, \quad (1)$$

where $\alpha_{\text{mlt}} \rightarrow A$ as $\beta \rightarrow -\infty$. Fig. 4 shows this relationship with an amplitude $A = 20$ and a base mixing length parameter $\alpha_{\text{mlt,base}} = 2.0$. The mixing length parameter increases sharply around $\beta = 0.3$ to target the region at the base of the convective envelope. This prevents the sudden drop in β , which is characteristic of the onset of the FeI, by increasing the efficiency of convective transport. The gradient of the slope has been chosen such that the mixing length parameter is unaffected for $\beta \gtrsim 0.7$. For the thermal pulse cycle shown in Fig. 3, this results in an increased mixing length parameter for the $\lesssim 0.01 M_\odot$ of mass above the core boundary. It is also worth noting that the shape of the β profile varies depending on the core and

envelope masses of the model. In particular, more massive cores result in lower values of β_{min} throughout the evolution.

While this increased mixing at the base of the convective envelope does not affect the envelope structure directly, it does change the efficiency of HBB in models that are massive enough ($M \gtrsim 4 M_\odot$). HBB occurs during the interpulse period but ceases when the envelope mass is sufficiently reduced. In contrast, the FeI occurs when a sufficient amount of envelope mass has been removed and during the thermal pulses. Thus, we increase the mixing length parameter during the luminosity peaks of thermal pulses using an increased amplitude when the hydrogen-burning luminosity, L_H , drops,

$$A_1 = \begin{cases} \alpha_{\text{mlt,base}} & \log(L_H/L_\odot) > 1 \\ 30 & \log(L_H/L_\odot) < 1. \end{cases} \quad (2)$$

We also increase the mixing length parameter during the interpulse period at the end of the TP-AGB to prepare the models for post-AGB evolution. This is done using an amplitude A_2 which increases linearly with the decreasing ratio of envelope mass to core mass, M_{env}/M_c , once the envelope mass is less than twice the core mass,

$$A_2 = \begin{cases} \alpha_{\text{mlt,base}} & M_{\text{env}}/M_c > 2 \\ -9M_{\text{env}}/M_c + 20 & M_{\text{env}}/M_c \leq 2. \end{cases} \quad (3)$$

The amplitude used in equation (1) is the maximum of those two amplitudes,

$$A = \max(A_1, A_2). \quad (4)$$

In Fig. 3, we compare the evolution of T_{eff} (panel a) and β_{min} (panel b) for a thermal pulse during which the FeI occurs without our routine (solid line), and with our routine to prevent it (dotted line). The model sequences deviate at the onset of the FeI as shown by the drop in β_{min} when our routine is not used. We have tested that the routine does not impact the efficiency of TDU by further increasing the value of A_1 , which we find has no impact. This method also avoids the FeI without increasing the efficiency of HBB or altering the surface properties. In fact, it has negligible impact on the stellar evolution other than to allow stars to evolve past the FeI. In addition, it works at all masses at solar metallicity and is easily adaptable to other metallicities by simply adjusting the strength of the mixing length parameter boosting through the amplitude A . In appendix E we detail the implementation of this method in MESA.

4.3 Other methods to prevent the FeI in massive and post-AGB models

The FeI also occurs in post-AGB models as detailed in Bertolami (2016). They found the instability leads to very short time-steps ($\Delta t \sim 1$ hour) and thus millions of time-steps are required to compute the post-AGB. In a similar vein to the methods previously discussed, they avoid the instability by forcing adiabatic convection. However, this is only at the very bottom of the convective envelope to still obtain the correct stellar radius and effective temperature. They tested a model with $M \sim 0.56 M_\odot$ with and without the alteration of the temperature profile and found no difference however models with larger masses were not compared. Oomen (2020) also boosted the convective-energy transport efficiency by reducing the temperature gradient at the base of the convective envelope. In $0.65 M_\odot$ post-AGB stars they could limit the impact of efficiency boosting on the effective temperature to several 100 K when they use a minimum temperature for boosting of $\log(T/\text{K}) > 4.9$. However at higher masses the effect on the stellar structure increases. Our increased mixing length parameter technique also effectively prevents the FeI in post-AGB

models. However, when the envelope mass is reduced, the Fe-opacity peak is closer to the stellar surface and thus any techniques to boost convective energy transport are more likely to affect the stellar radius and effective temperature. While our technique does not impact the stellar structure in our AGB models, it has not been thoroughly tested on the post-AGB and may change the stellar structure when β drops very close to the surface.

A similar situation to the FeI occurs in the envelopes of massive stars near the Eddington limit. To aid the computation of such models in MESA, a treatment of convection known as MLT++ was developed by Paxton et al. (2013) and updated in Jermyn et al. (2022). This treatment reduces the superadiabaticity in some radiation-dominated convective regions much like the method used by Bertolami (2016). Jermyn et al. (2022) discuss that this enhancement in the efficiency of convective energy transport is supported by results from 3D simulations (Jiang et al. 2015; Schultz, Bildsten & Jiang 2020), although the method is not calibrated to detailed simulations or observations. The enhanced convective energy transport can inhibit the formation of density inversions to facilitate calculations, much like our increased mixing length parameter routine. However, we find using MLT++ in TP-(S)AGB envelopes leads to peculiar surface properties during thermal pulses and thus we find our targeted approach much more suitable for countering the FeI.

5 DISCUSSION

TP-(S)AGB instabilities have long plagued models produced by 1D stellar evolution codes. While the onset of convergence problems differs depending on the carbon enrichment of the envelope, which lowers the surface temperature, and chosen physics, such as the mixing length parameter, which compresses the envelope at larger values, the occurrence of instabilities is inevitable at low effective temperatures without numerical methods to prevent them. To illustrate this, we focus on the four model sets that were compared in Rees et al. (2024) and are also computed at $Z = 0.014$. The MONASH models (Karakas 2014; Karakas & Lugaro 2016), which have similar TDU efficiencies to our MESA models, terminate due to convergence issues with $\approx 1 M_\odot$ of envelope remaining at intermediate initial masses ($M_i \gtrsim 2.25 M_\odot$). The Frascati RAphsonNewton Evolutionary Code (FRANEC) Repository of Updated Isotopic Tables & Yields (FRUITY) models (Cristallo et al. 2009, 2011, 2015) extrapolate past convergence problems to compute the remainder of the TP-AGB. The total mass at the last computed thermal pulse is $1.38 M_\odot$ in their $M_i = 2 M_\odot$ model sequence and $2.24 M_\odot$ in their $M_i = 6 M_\odot$ model sequence. Ventura et al. (2018) do not mention convergence issues in their TP-(S)AGB models produced with the ATON code; however, their surface abundance plots in fig. 10 indicate that the models terminate early at stellar mass ranging from $\sim 1 M_\odot$ for an $M_i = 2 M_\odot$ sequence to $\sim 2 M_\odot$ for $M_i \gtrsim 6 M_\odot$ sequences.

Calculations of the TDU efficiency at small envelope masses have been limited by the ability to converge such stellar structures through thermal pulses. In Fig. 5, we plot the envelope mass (panel a), hydrogen-exhausted core mass (panel b), and radius (panel c) of an $M_i = 4 M_\odot$ model evolved until there is only $0.01 M_\odot$ of envelope mass remaining. For comparison, we also compute the evolution without our routines to prevent the HRI and FeI, which shows that our routines do not impact the TP-AGB evolution, other than by allowing it to continue. We find that TDU episodes decrease in efficiency when $M_{\text{env}} \lesssim 1.0 M_\odot$, although only by a modest amount. Over the last five thermal pulses, the TDU efficiency decreases from $\lambda \sim 1$ to $\lambda \sim 0.85$ during the last pulse with $M_{\text{env}} = 0.09$.

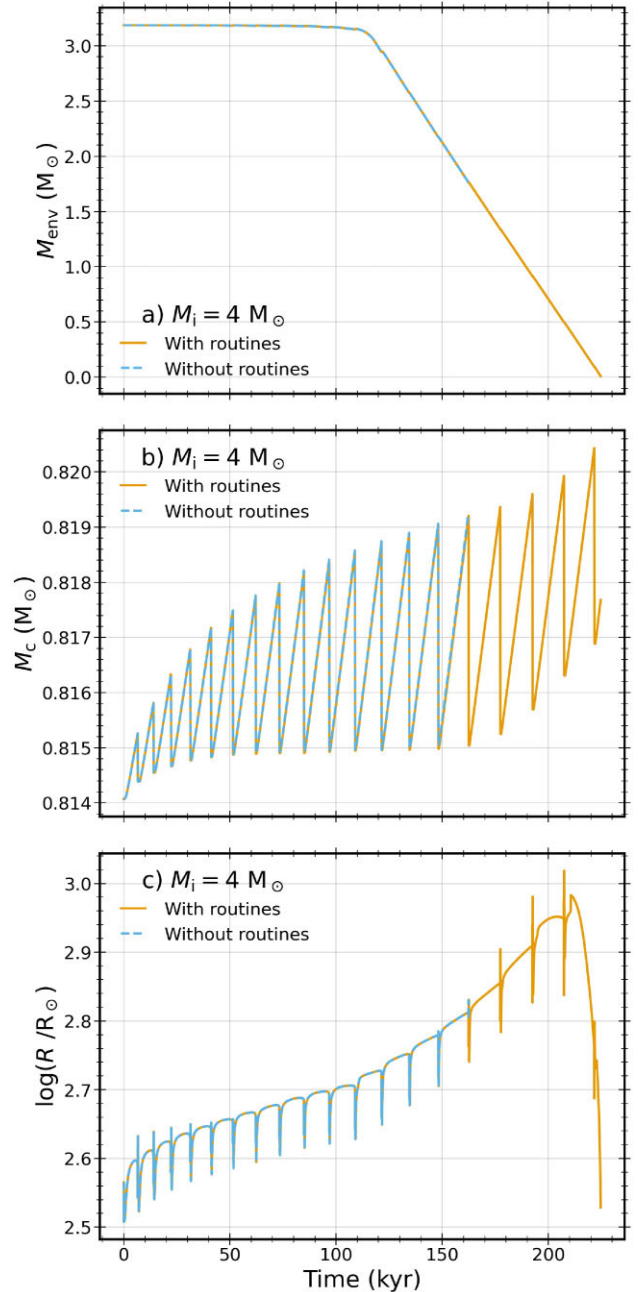


Figure 5. The envelope mass, M_{env} (a), core mass, M_c (b), and stellar radius, R (c), during the TP-AGB phase of an $M_i = 4 M_\odot$ model star. Without the routines developed in Sections 3.2 and 4.2, convergence fails during the 18th thermal pulse (black dashed line) due to the HRI and FeI.

However, due to the decreased TDU efficiency, the core mass does grow more rapidly as the envelope is removed. Thus, the evolution at low envelope masses is very important to determine the final mass of the WD remnant (Marigo et al. 2020; Marigo 2022). We also see that the third- and second-to-last thermal pulses result in large radial expansions, reaching $\log(R/R_\odot) \approx 3.02$, which could have important consequences for binary star interactions in systems with a short enough orbital period.

Unfortunately, we cannot directly check the impact of our routines on the TDU efficiency during the last five thermal pulses because we cannot converge through the TDU events without them. However,

given that our routines have no impact on the TDU efficiency previous to this, we assume that this does not change. Furthermore, if the routine to avoid the FeI has an impact on the TDU, then we should find a dependence of the TDU efficiency on the amplitude of the mixing length parameter boosting, which we do not. In addition, if the routine to prevent the HRI has an impact on the TDU, then this should be a stronger effect at lower effective temperatures. It follows that the TDU efficiency would decrease and then increase again for the last thermal pulse as the effective temperature does the same. However, we find a smooth decrease in the TDU efficiency as the envelope mass decreases. We thus conclude that the modest decrease in TDU efficiency is due to the reduced envelope mass, and not our routines.

We also compute the entire TP-AGB evolution for an $M_i = 5 M_\odot$ star, which is massive enough to undergo HBB (Fig. 6). We find only a very small decrease in TDU efficiency as the envelope is removed, with λ decreasing from a maximum of 0.97 to 0.92 for the last thermal pulse with $M_{\text{env}} = 0.2 M_\odot$. Our results seem to indicate that the larger the core mass, the less significant the decrease in TDU efficiency as the envelope mass is decreased. In Fig. 6(c), we show the evolution of the surface C/O ratio. At the start of the TP-AGB evolution, the ratio is approximately constant, while TDU is weak and the temperature at the base of the convective envelope is too cool for HBB. After ≈ 40 kyr, HBB starts to convert carbon into nitrogen, which decreases the C/O ratio and keeps it low, even while the TDU episodes become more efficient. Once the envelope mass has been sufficiently reduced ($M_{\text{env}} \approx 2 M_\odot$), after ≈ 125 kyr, HBB ceases and TDU dominates the C/O ratio by enriching the envelope in carbon. The enrichment gets stronger as the envelope mass is reduced and the carbon mixed into the envelope is diluted by less material. For the last three thermal pulses, C/O exceeds 1 and the star would be observed as a carbon star. This carbon-rich phase at the end of the TP-AGB lasts ≈ 20 kyr, which is ≈ 10 per cent of the total thermally pulsing lifetime. Such stars would be observed as high-luminosity carbon stars but would be more rare than low-luminosity carbon stars that have increased lifetimes. Izzard & Tout (2004) modelled the carbon star luminosity function of the Large Magellanic Cloud and found that the high-luminosity tail was not well reproduced. This could be, in part, due to the assumed TDU efficiency at low envelope masses. A detailed review of the properties of TP-(S)AGB stars over a range of core masses and envelope masses will be conducted in a future work. In particular, the new models will be used to update the synthetic AGB model developed in Izzard et al. (2004) for use in the population synthesis code `binary_c` (Izzard et al. 2006, 2009, 2018).

As discussed in Henry et al. (2018), computing the last TDU event(s) is also important for predicting the abundances of planetary nebula. They highlight that even if the TDU efficiency is reduced, significant changes in the surface abundances can still happen due to the reduced envelope mass in which the dredged-up material is diluted in. They find this to be an important difference between the MONASH, FRUITY, and ATON models, which do not reach the post-AGB and the La Plata stellar evolution code (LPCODE) models (Bertolami 2016) that are evolved up to WDs.

Finally, we discuss the validity of our approach of preventing instabilities from occurring and continuing to compute the evolution. We believe that the HRI is a consequence of attempting to model an envelope that is unstable to pulsations using a 1D hydrostatic code. Thus, it would not occur in real stars, in which hydrogen recombination drives envelope pulsations instead. However, it is thought that the FeI may be a real instability that leads to a significant mass ejection event (Lau et al. 2012). Due to the large amplitude pulsations

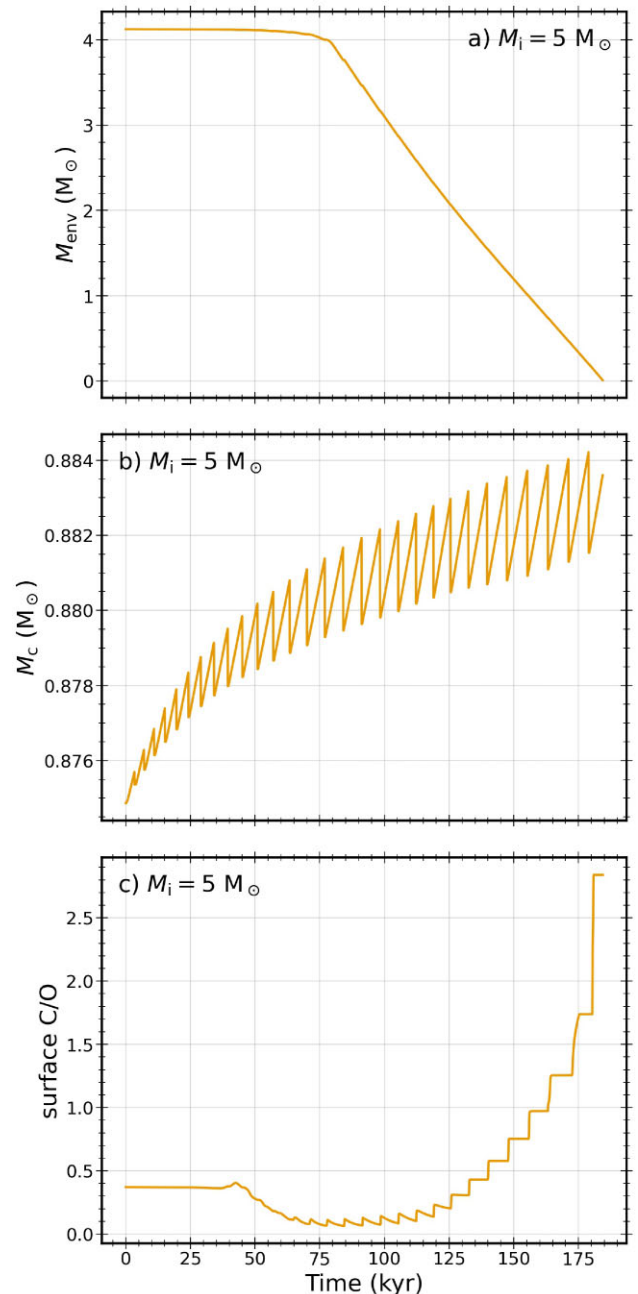


Figure 6. The envelope mass, M_{env} (a), core mass, M_c (b), and surface C/O ratio (c) during the TP-AGB phase of an $M_i = 5 M_\odot$ model star.

that occur in AGB stars at the end of the TP-AGB, we cannot use 1D hydrostatic models to accurately model envelope properties or mass-loss rates. Instead, we would require 3D hydrodynamical codes to constrain the behaviour of AGB envelopes. Similarly, any real instabilities would manifest differently in a 3D hydrodynamical model. Thus, we take the approach of assuming a mass-loss rate from an estimate derived from observations (Vassiliadis & Wood 1993) and preventing instabilities that terminate the evolution from occurring. When more accurate mass-loss rates become available from 3D hydro models, such prescriptions can be applied as we are able to compute 1D hydrostatic models that do not terminate, regardless of the mass-loss rate chosen. If it is found that the FeI does lead to significant mass ejection events, these can also be applied. In the meantime, we

are able to constrain the behaviour of thermal pulses at low envelope masses using our models.

6 CONCLUSIONS

The HRI and FeI usually occur during the luminosity peak following a thermal pulse and so, without inspection, it is easy to conflate the two. The FeI can be diagnosed by a sudden drop in the minimum value of the ratio of gas to total pressure, $\beta_{\min}$. The HRI may manifest in different ways depending on the temporal resolution. However, a sudden change in surface properties during the luminosity peak, which is not accompanied by a sudden drop in $\beta_{\min}$, is likely to be the HRI. We find that the HRI occurs before the FeI, although this depends on the time resolution. The HRI can be prevented by damping the energy released by hydrogen recombination in the outer envelope. Outside of an instability, these energy contributions are negligible compared to the energy released by the burning shells and thus our damping does not impact the stellar structure. The FeI can be prevented by boosting the energy transport at the base of the convective envelope using an increased mixing length parameter. This is only done during the luminosity peak of thermal pulses and during the interpulse when the envelope is sufficiently reduced to avoid impacting the efficiency of HBB in intermediate-mass stars. With our routines to prevent these instabilities, TP-(S)AGB models can now be produced up to the post-AGB for a range of physical assumptions and without human intervention. Thus, the properties of TP-(S)AGB stars with small envelope masses can be studied with the 1D stellar evolution code MESA.

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DATA AVAILABILITY

MESA is open source software available at <https://docs.mesastar.org/en/release-r23.05.1/>. The `inlists` and `run_star_extras` code used will be available at the MESA marketplace (https://cococubed.com/mesa_market/inlists.html) upon publication. The MESA output data will be available at https://zenodo.org/communities/binary_c-community upon publication.

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APPENDIX A: MESA CONTROLS FOR CONVERGENCE

The He-shell flash can be a tricky phase for stellar evolution codes to evolve through and convergence issues occur frequently, particularly at $M_i \gtrsim 2 M_\odot$. We find some MESA controls are necessary to ensure convergence over a range of initial masses. There are two required controls relating to the temperature gradient equation. During the He-shell flash, the model sometimes fails to converge as it struggles to find an acceptable solution for the temperature gradient equation in the intershell region just above the CO core. In MESA, the standard way of calculating the temperature gradient equation, dT/dm , is by using the hydrostatic dP/dm . However, in dynamic situations the use of hydrostatic dP/dm is inappropriate and dP_{rad}/dm should be used instead. While for the majority of AGB evolution the hydrostatic approximation is very good, at the start of thermal pulses the star contracts on a much shorter time-scale and this poses a

problem for a hydrostatic solution. We allow use of the dynamic form using the `use_dPrad_dm_form_of_T_gradient_eqn = .true. control`.

The second necessary control is `convergence_ignore_equL_residuals = .true.` to ensure the residuals of the temperature gradient equation are ignored and thus do not drive down the time-step. Neither of these controls have any impact on the evolution other than preventing termination from convergence problems and so are used for all models.

APPENDIX B: RESOLVING THE THIRD DREDGE-UP

In Rees et al. (2024), we showed that it was crucial to use sufficient time resolution during TDU, otherwise the efficiency can be significantly underestimated (or overestimated for SAGBs). We altered the time resolution by using the time-step control `varcontrol_target`, due to the lack of an existing time-step control that corresponds to changes in the bottom of the convective envelope. However, the MESA documentation does not recommend performing time resolution studies using `varcontrol_target` due to its general nature. It is preferable instead to use specific controls to target the regime being studied. Bearing this in mind, we test an alternative method to resolve the TDU by reducing the time-step if the change in the mass coordinate at the bottom of the convective envelope, Δm_{bce} , decreases by more than some allowed amount. Because the thickness in mass of the intershell region is strongly dependent on core mass, we set the allowed amount to be some factor, f , of the growth of the hydrogen-exhausted core mass during the preceding interpulse period. Thus, for example if $f = 0.001$, approximately 1000 time-steps would be needed to resolve a TDU event with efficiency, $\lambda = 1.0$. This routine is implemented using the `other_timestep_limit` hook.

Upon testing our alternative method, we find that as the stellar mass decreases, the time resolution required to correctly resolve the TDU efficiency increases. For example, when resolving the TDU in an $M_i = 4 M_\odot$ star, a factor of $f = 0.0005$ (~ 2000 time-steps) is sufficient while $M \sim 4 M_\odot$ at the start of the TP-AGB. However, when the stellar mass is reduced to $M \sim 1.8 M_\odot$, a factor of $f = 0.00005$ (~ 20000 time-steps) is necessary to resolve the TDU efficiency. Thus, to resolve the TDU throughout the TP-AGB evolution, the factor f must be decreased as the envelope mass decreases. Such a calibration may be tricky as it likely also depends on core mass. We note however, that our first method to resolve the TDU using `varcontrol_target` achieves an increase in time resolution with decreasing envelope mass by using a constant value of the `varcontrol_target` during TDU, $v_{i,\text{TDU}} = 2 \times 10^{-5}$. Because of this easier implementation, we would still recommend use of the first method, however both are available in our `run_star_extras` code.

Previous works have given contradictory results for the efficiency of TDU at reduced envelope mass. Straniero et al. (1997) found that TDU vanished when $M_{\text{env}} \lesssim 0.5 M_\odot$ for the initial mass range $1 \geq M \geq 3 M_\odot$ at solar metallicity. On the other hand, Stancliffe & Jeffery (2007) found no decrease in TDU efficiency in a $M_i = 1.5 M_\odot$ model sequence. The MONASH (Karakas 2014; Karakas & Lugaro 2016) and FRUITY (Cristallo et al. 2009, 2011, 2015) models both show a decrease in TDU efficiency as the stellar mass decreases, although the actual dependence between the two differs (Rees et al. 2024, fig. 6). For example, TDU ends in the FRUITY $M_i = 6 M_\odot$ model sequence when the stellar mass is reduced to $\sim 2 M_\odot$; however, the equivalent MONASH model sequence shows

no decrease in TDU efficiency up to its termination at $M \sim 2 M_{\odot}$. It is crucial to point out that at low envelope masses, we find the TDU efficiency strongly depends on the time resolution used. Our MESA models require tens of thousands of time-steps to resolve the TDU at reduced envelope mass, compared to thousands when $M \sim M_i$. We thus advise that all future studies should be careful that the TDU is being properly resolved at low envelope masses. However, differences in TDU efficiency between sets of models are likely also caused by differences in the chosen physics and/or stellar evolution codes.

APPENDIX C: MESA ENERGY EQUATION

MESA has two different options for the energy equation: the `eps_grav` form and the `dedt` form (Paxton et al. 2018, section 8). Under degenerate conditions, the `eps_grav` form is generally found to be preferable because errors generated from time discretization and equation-of-state inconsistencies manifest in energy errors rather than entropy errors when the `dedt` form is used. Given TP-(S)AGB stars have degenerate cores, we opt for the `eps_grav` form of the energy equation that we find to be more reliable during the He-shell flashes of thermal pulses. The quantity `eps_grav` is defined as,

$$\epsilon_{\text{grav}} = -\frac{De}{Dt} - P \frac{DV_{\rho}}{Dt}, \quad (\text{C1})$$

where e is the specific internal energy, P is the pressure, V_{ρ} is the specific volume, and D/Dt is the Lagrangian time derivative.

APPENDIX D: IMPLEMENTATION OF HYDROGEN RECOMBINATION INSTABILITY PREVENTION

As discussed in Section 3.2, the HRI can be prevented by damping out energy injections from hydrogen recombination in the

outer envelope. In MESA, the ionization/recombination energy is included in the ϵ_{grav} term of the energy equation. Thus, we use the `other_eps_grav` hook to adjust the value of ϵ_{grav} . Note that this routine is called after the standard `eps_grav` calculation so that we know the true profile of this quantity. From Ivanova (2018, fig. 1), we see that hydrogen recombination occurs at temperatures $\log(T/K) \lesssim 4.5$ in AGB envelopes. In the range $4.2 < \log(T/K) < 4.5$, we reduce ϵ_{grav} by multiplying it by a constant that linearly reduces from 1 at $\log(T/K) = 4.5$ to 0 at $\log(T/K) = 4.2$. In the remaining outer envelope at $\log(T/K) < 4.2$, we also set $\epsilon_{\text{grav}} = 0$. The routine is implemented in `agb_routines.inc` in the `src` folder that is provided alongside the `inlist` files.

APPENDIX E: IMPLEMENTATION OF FE-PEAK INSTABILITY PREVENTION

To implement our variable mixing length parameter routine, we use the `other_alpha_mlt` hook. This hook sets the mixing length parameter in an array called `alpha_mlt` while the global value set in the `inlist` is stored in the scalar `mixing_length_alpha`. However, there is a bug in MESA 23.05.1, which causes the code to only use the global value `mixing_length_alpha` in convection calculations. This can be fixed by changing `s% mixing_length_alpha` to `s% alpha_mlt(k)` on line 120 of `star/private/turb_info.f90`. We then implement our variable mixing length parameter using equations (1)–(4). The routine is implemented in `agb_routines.inc` in the `src` folder that is provided alongside the `inlist` files.

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